communications earth & environment | Article

A Nature Portfolio journal

https://doi.org/10.1038/s43247-025-02169-5

Complexities of the global plastics supply chain revealed in a trade-linked material flow analysis

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Plastic has become an integral part of our lives. Addressing the global environmental concerns of plastics requires a comprehensive analysis along the whole supply chain. Here, we provide a global trade-linked material flow analysis of plastics for the year 2022. Globally, 436.66 million tonnes (Mt) of plastics were traded in 2022, with final products alone accounting for 111 Mt. Our findings suggest that plastics are subject to geographical concentration, with feedstocks concentrated in oil-resource rich countries and processing in countries with large manufacturing capacity. Furthermore, there is a significant shift in waste disposal: incineration is emerging as a prominent waste disposal method (34%), landfill is decreasing substantially (40%), while the global recycling rate remained stagnant (9%). Income disparities among countries diminish in the context of plastic waste imports, reshaping the global plastics trade patterns. Uncovering the complex plastic supply chain is crucial for reducing pollution and promoting sustainable plastic management.

Plastics are one of the most ubiquitous materials in our planet, owing to their versatility, durability, and relatively low cost. The global demand for plastics has quadrupled over the past decades[1] and it is projected to double by 2050[2], resulting in severe impacts on the environment and human health[3,4]. The global annual production of plastics grew from 2 Mt in 1950 to 400 Mt in 2022[5], at an annual growth rate of 8.4%.

Plastic pollution is a pressing global issue that poses significant environmental, economic, and public health challenges. Intergovernmental conventions have emerged as pivotal instruments in the fight against plastic pollution, aiming to coordinate international efforts, set common goals, and establish frameworks for sustainable plastic waste management. In March 2022, the United Nations Environment Assembly under the United Nations Environment Programme (UNEP) in Nairobi unanimously approved a resolution to create a legally binding treaty by 2024 to prevent, reduce, and end global plastic pollution[6]. The resolution and the subsequently intergovernmental negotiations are aimed to address the full lifecycle of plastic, including its production, design and disposal. Despite the growing recognition of plastic pollution as a critical global environmental issue, there is a notable lack of comprehensive analysis of plastics along their supply chain.

Addressing the environmental impacts of the global plastics production and consumption require a deeper understanding of the complex supply chain of plastic production, use, application and disposal. Material flow analysis (MFA) is an effective method for studying the metabolism of materials and resources[7]. To date, the global stocks and flows of plastics have not been comprehensively evaluated along the whole supply chain. Geyer et al.[1] presented a first global material flow analysis of plastics from 1950 to 2015, by developing and combining data on plastic production, use and end-of-life. While their study provides a significant advance in quantifying global plastic production, use, and disposal, their temporal scope may not fully account for rapid technological advancements and changes in plastic use patterns in the recent years. In addition, by focusing on global plastics pattern, their findings may not capture the regional variations in production, consumption, and waste management practices.

Existing studies have focused on regional and national plastics material flow analysis, including Europe[8–11], China[12–16], USA[17,18], Africa[19,20], Middle East[21], India[22,23] and Japan[24]. Other national studies were conducted in United Kingdom[25], South Africa[26] and Thailand[27]. The call to end plastic pollution resulted in a wide number of studies that have evaluated global plastic waste flows and disposal[28–33]. The majority of these studies have provided an MFA at national or regional level, however a comprehensive analysis of global plastics stocks and flows along the whole supply chain has not been examined.

This study addresses the exiting research gap by providing a first global trade-linked material flow analysis of plastics based on the year 2022. First, the global flows and stocks of plastics produced in 2022 will be uncovered with a detailed breakdown per polymer types and plastics application sectors. Second, the trade of plastic products along the whole supply chain will be evaluated from feedstock to plastic waste products. Third, the global waste disposal

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and end-of-life fates will be analyzed by providing a regional comparison of plastic waste treatment and disposal. Finally, based on the findings policy recommendations will be proposed for a global sustainable management of plastic resources. The findings will ultimately support the treaty negotiations to enhance plastics resource efficiency and reduce plastics pollution.

Results

Global trade-linked plastics material flow analysis

The global plastic production in 2022 accounted of 400 Mt, from which 362 Mt are produced as plastic virgin resin production and around 38 Mt generated as secondary plastics from plastic mechanical recycling (Fig. 1). Around 98% of the global virgin plastics produced in 2022 is generated from fossil-fuel based feedstocks (44% derived from coal, 40% from petroleum, 8% from natural gas, 5% from coke and 1% from other sources). Only 2% of the global plastics feedstocks are generated from bio sources. Losses from virgin plastics production stage accounted for 13.42 Mt. A total of 348.52 Mt of virgin plastics were processed, and the total amount of plastics manufactured in 2022 was around 386.41 Mt (comprising the recycled plastic diverted in the manufacturing stage). The total plastics losses in the manufacturing stage accounted for around 4.24 Mt. A total of 382.12 Mt of plastics entered the use stage, with 158.04 Mt in packaging, 72.05 Mt in building and construction, 32.02 Mt in automotive, 28.02 Mt in electrical and electronics, 28.01 Mt in household and textile, 16.01 Mt in agriculture, and 48.03 Mt in others.

We estimate plastics stocks in different plastics industrial sectors based on the global average product life-time. The total amount of plastics stocks amounted to ~115 Mt, with the building and construction sector accounting with the largest share (50%), followed by automotive, household and textile, electrical and electronics with 18%, 13%, and 9%, respectively. The packaging sector contributed 2 kt to the total stocks. Similarly, the agricultural sector accounted for a relatively small share, with 1.30 Mt, representing about 1% of global plastics stocks.

The total waste generated in 2022 amounted to 267.68 Mt of which 74.75 Mt were sorted and collected for mechanical recycling and 6.66 Mt traded. Among the total plastic waste collected and sorted, 37.96 Mt of plastics have been recycled (accounting for 9% of the primary production), 30.66 Mt were diverted to incineration facilities, and the remaining 6.25 Mt landfilled. In total 29.60 Mt of plastics have been mismanaged, 103.10 Mt landfilled, 89.99 Mt incinerated. A total of 436.66 Mt of plastics were traded in 2022 (Fig. 2). Among those, feedstocks trade accounted for 71 Mt (Fig. 2a). EU28 was the largest exporter of plastics feedstocks (31%), followed by Oth Asia (26%), and USA (14%). The largest importers of feedstocks for plastics production are China (31%), followed by EU28 (30%), and ROW (14%) (for the regional denomination and countries classification see Supplementary Table 10). EU28's large amount of import and export is due to the trade of plastics within the region and the re-export of feedstock regionally. A total of 30 Mt additives were traded (Fig. 2b), with China as the largest exporter (23%) followed by EU28 (22%), Oth Asia (18%), USA (16%), and Middle East (13%). The largest importers of additives were China, followed by EU28, Oth Asia, USA, Middle East. Around 152 Mt of plastics were exported as primary plastics (Fig. 2c). Oth Asia region dominates as the largest exporters of primary plastics accounting with 51.55 Mt (32%), followed by EU28 (44.17 Mt) and USA (20.34 Mt).

Intermediate stage has been divided between intermediate forms of plastics and intermediate manufactured with a total trade of 47 Mt and 19 Mt respectively (Fig. 2d, e). Intermediate forms of plastic products refer to plastic products that are still at an early stage of the supply chain. Such forms of plastic products consist of primary plastics that have already been processed and assembled into larger elements, such as sheets, films, plates, and yarns. Intermediate forms of plastics will then be further molded, shaped, manufactured, assembled to produce intermediate and final manufactured products. China is the largest exporter of intermediate forms of plastics (14.21 Mt) and intermediate manufactured plastics (9.44 Mt). Similarly, EU28 was responsible for 15.07 Mt of intermediate forms of plastics and for 1.73 Mt for intermediate manufactured plastics, Oth Asia for 5.73 Mt and 1.42 Mt respectively, India for 1.59 Mt and 1.22 Mt respectively, and Japan for 0.86 Mt and 0.24 Mt respectively.

Final plastics products accounted for the largest trade share along the whole supply chain of plastics with a total of 111 Mt (Fig. 2f). China, as one the countries with the largest manufacturing capacity exported a total of 44.25 Mt (45%) followed by Oth Asia region (21%), and EU28 (18%). The largest importer of plastics final products was EU28 (35%), followed by USA (20%), Oth Asia (14%), ROW (13%), China and Middle East (5% each), Africa (4%), Japan (3%) and India (2%).

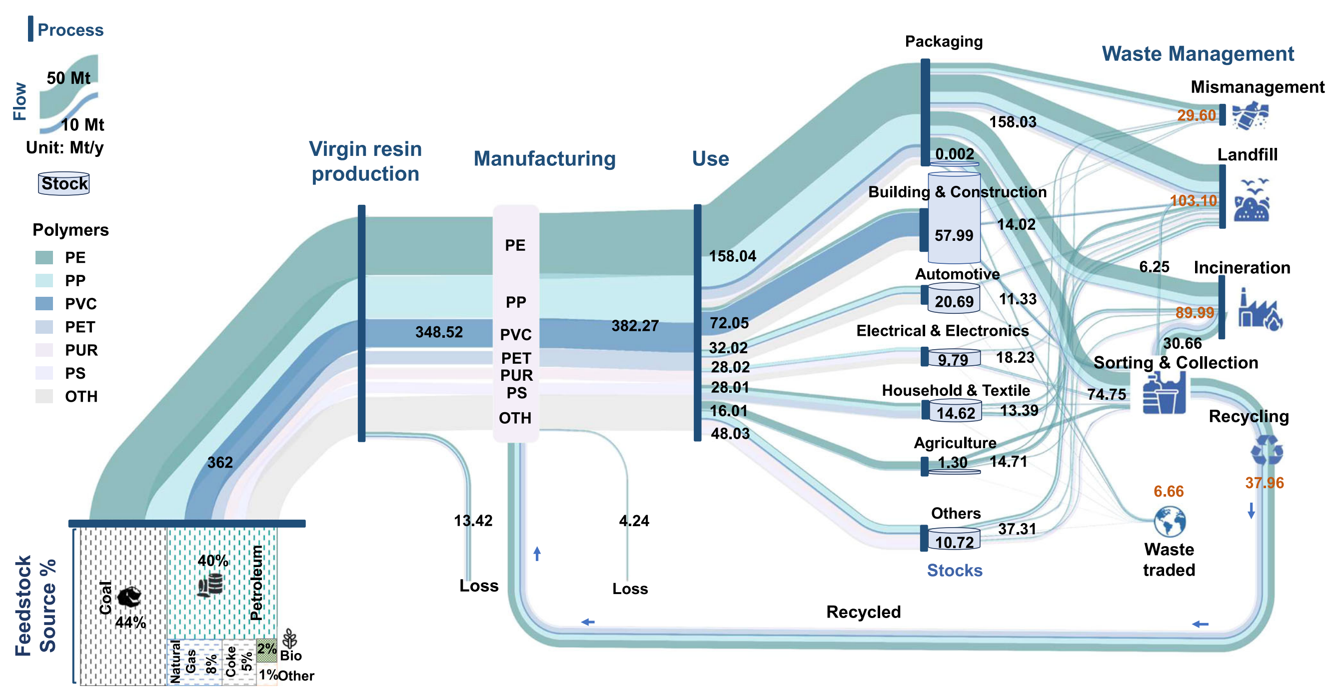

Fig. 1 | Global plastic cycles in 2022. The global plastics cycles are illustrated by flows. Color and width of flows correspond to the types and mass of plastics, respectively. All units in Mt/y. *Note: minor discrepancies in values are due to rounding during mass balance calculations.*

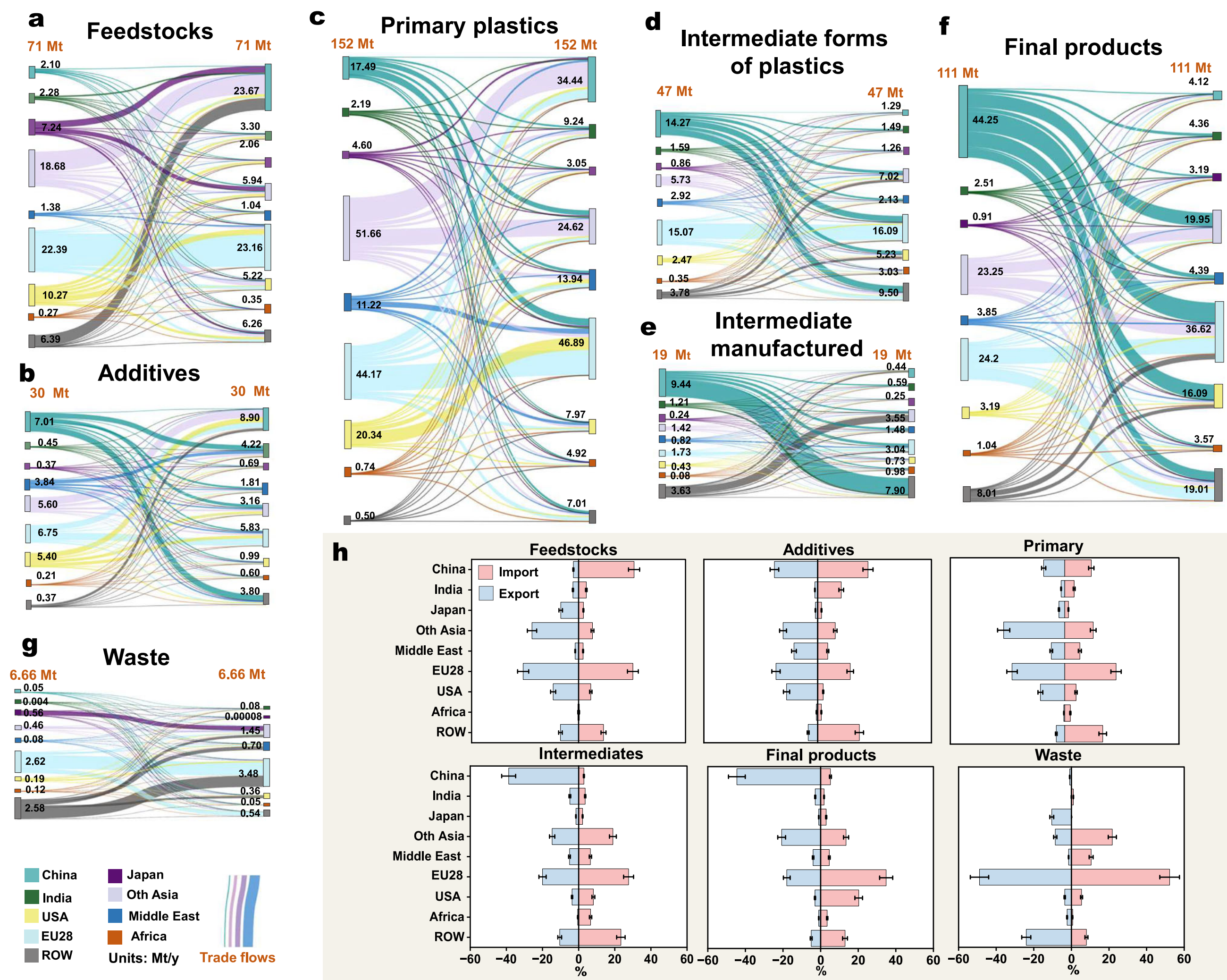

Fig. 2 | Global plastics trade along the whole supply chain. The global trade of each plastic supply chain category is mapped as follow: **a** Feedstocks trade flows. **b** Plastics additives trade flows. **c** Primary plastics trade flows. **d** Intermediate forms of plastics trade flows. **e** Intermediate manufactured plastics trade flows. **f** Final plastics trade flows. **g** Plastics waste trade flows. **h** The global import and export rates along the supply chain of plastics. The global mapped trade values are expressed in Mt/y. Uncertainty analysis results are shown through error bars. A detailed analysis of plastics composition for each trade category is illustrated in Supplementary Figs. 3–9. Regional classification is provided in Supplementary Table 10. *Note: minor discrepancies in values are due to rounding during mass balance calculations.*

The global plastic waste traded in 2022 accounted to 6.66 Mt (Fig. 2g). EU28 exported 49% and imported more than 52% of the global plastic waste. The second largest exporter of plastic waste was Japan (10%), followed by Oth Asia region (9%), USA (4%), Africa (2%) and ROW (24%). China's ban on plastics waste import in 2018 changed the global trend of plastic waste import and export. China had been the largest importer of plastic waste for 25 years[34]. In 2022, the largest importer of plastics waste was EU28 (3.48 Mt), followed by Oth Asia (1.45 Mt), Middle East (0.7 Mt) and USA (0.36 Mt). EU28 become a net importer of plastic waste in 2022 with its total waste import accounting for 3.48 Mt while the total export 2.62 Mt. Netherland imported a total of 0.81 Mt of plastic waste, accounting for 23% of the total import at regional level and 12% at a global level. Germany was responsible of 10% of the total import of plastic waste in EU28, followed by Belgium (9%) and Austria (8%) (Supplementary Fig. 10a). These European countries alone accounted for around 50% of the total plastic waste import in EU28. Regarding plastic waste export, our results show that Germany has the highest export rate of plastic waste in EU28 accounting for 0.49 Mt (17% of regional export), followed by Belgium (15%), France (13%) and Italy (10%) (Supplementary Fig. 10b). Oth Asia countries continue to dominate the global plastics trade market as recipient countries. Malaysia with a total waste import of 0.35 Mt registered the largest plastic import share in the region (24%), followed by Viet Nam (21%), Indonesia (13%), and Thailand (12%) (Supplementary Fig. 11). The global import and export rate for plastics along different stages of the supply chain is illustrated in Fig. 2h. China, EU28 and Oth Asia countries dominate the largest imports and exports in different stages, emerging as major players within the global supply chain of plastics. An analysis of the plastic component of each plastic stage considered in the study is illustrated from Supplementary Figs. 3– 9.

Global and regional plastics production and consumption

The global production of plastics accounted for 400 Mt in 2022. China was the largest producer of plastics (32%), followed by Oth Asia (15%), USA (14%), EU28 (14%), Middle East (5%), India (5%), Africa (4%), Japan (3%), and ROW (8%) (Fig. 3a). The largest polymer produced worldwide is PE accounting of 26% of the global production. Other largely produced polymers are PP (19%), PVC (13%), PET (2%), PUR (5%) and PS (5%). Secondary plastics accounted for 9% of the global production and bio-plastics a relatively small fraction (0.5%). The regional polymer production with a breakdown per polymer type is illustrated in Supplementary Fig. 2.

In terms of consumption, China was the largest consumer of polymers with 20% of the global consumption. The second largest consumer of plastics is USA (18%), followed by EU28 (16%) and Oth Asia region (12%).

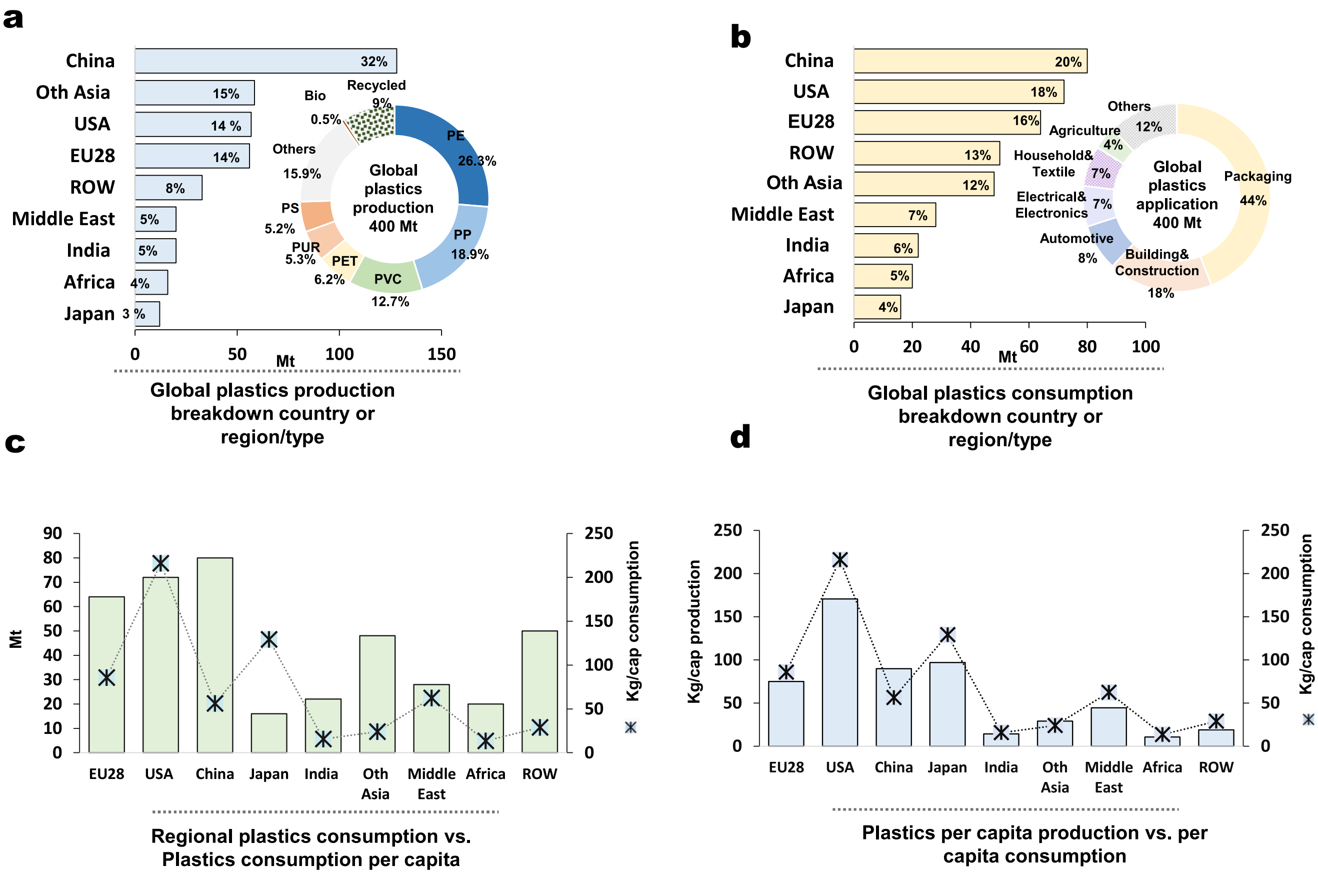

Fig. 3 | Global plastics production and consumption trends. **a** Global plastic production breakdown per country and polymer type. **b** Global plastic consumption breakdown per country and region and global plastic applications. **c** Regional plastics consumption amount and plastics consumption per capita (plastics consumption per region/country is expressed in Mt/y and plastics consumption per capita in kg/cap). **d** Plastics production per capita compared to plastics consumption per capita (values in kg/cap). For further details see Supplementary Table 1.

India and Japan accounted for 6% and 4% respectively, while Middle East and Africa for 7% and 5% of the global consumption (Fig. 3b).

In terms of consumption per capita, USA has the highest plastics consumption per capita (216 kg/cap). Plastics account for approximately 12% of USA municipal solid waste generation[35], with plastic containers and packaging being the main source of plastic waste. EU28 and Japan have also registered a relatively high plastic consumption per capita, with 86.6 kg/cap and 129 kg/cap respectively (Fig. 3c). According to the results, in terms of total quantity by weight, there is a significant degree of match between the geographic distribution of plastic production and consumption, thereby suggesting a spatial coupling of production-consumption patterns. However, when comparisons are adjusted on the basis of per capita consumption, disparities become more evident (Fig. 3d); regions with lower populations but substantial production capacities exhibit higher per capita production figures, contrasting with densely populated regions, but per capita consumption may be disproportionately high compared with local production. Such correlation suggests that local plastic production largely drives the consumption patterns within the same regions, potentially due to factors like logistical efficiency, economic policies, and market dynamics.

Global and regional plastics waste management

We estimate a total of 267.68 Mt of plastics waste was generated in 2022 (Fig. 4b). China is responsible of the largest regional waste generated accounting for 81.5 Mt (30%) of global waste. USA generated a total of 40.1 Mt of plastic waste, followed by Oth Asia region (35 Mt), EU28 (30 Mt) and India (9.48 Mt). Japan, Middle East and Africa generated a relatively small amount of plastic waste in comparison with 4.5 Mt, 1.63 Mt and 2 Mt respectively. ROW generated the remaining 63.47 Mt of the total global waste of plastics (Fig. 4c). The amount of the total waste generated includes the total waste/scrap imported and exported globally (6.6 Mt), which is further processed in recycling facilities in host countries or re-exported within the same region.

There is a wide diversity in technical advancement and plastic waste treatment system structures varying significantly worldwide. Globally, landfill remains the main destination of plastics waste, accounting for 103.37 Mt (40%). This amount shows improvement from the historical accumulated waste in landfill estimated by Geyer et al.[1] (79% of the total waste). This reflects China changes patterns in urban and rural solid waste management in the recent years. Our result showed that China incineration rate amounted to 60% (49.13 Mt) in 2022, while the total plastic waste landfill was only 14% (11.61 Mt) and plastics waste mismanaged accounted for 2% (1.71 Mt). China major shift towards air-controlled incineration of waste management is backed by the ambitious goal set by the country to achieve "zero landfill" of primary municipal solid waste by 2023[36]. Japan has one of the highest air-controlled incineration rates in the world, with 3.15 Mt (70%) of the total waste incinerated, 8% landfilled, and nearly 20% of the total waste recycled. Other developed countries made also major efforts in implementing strategic schemes for plastics waste management disposal. Plastic waste disposed in landfill in EU28 accounted for 8.7 Mt (29%), while mismanagement and incineration accounted for 3% and 38% respectively.

The total plastic waste landfilled in USA remains relatively high, with 30.47 Mt (76%) landfilled, 12% incinerated, 4% mismanaged and only 5%

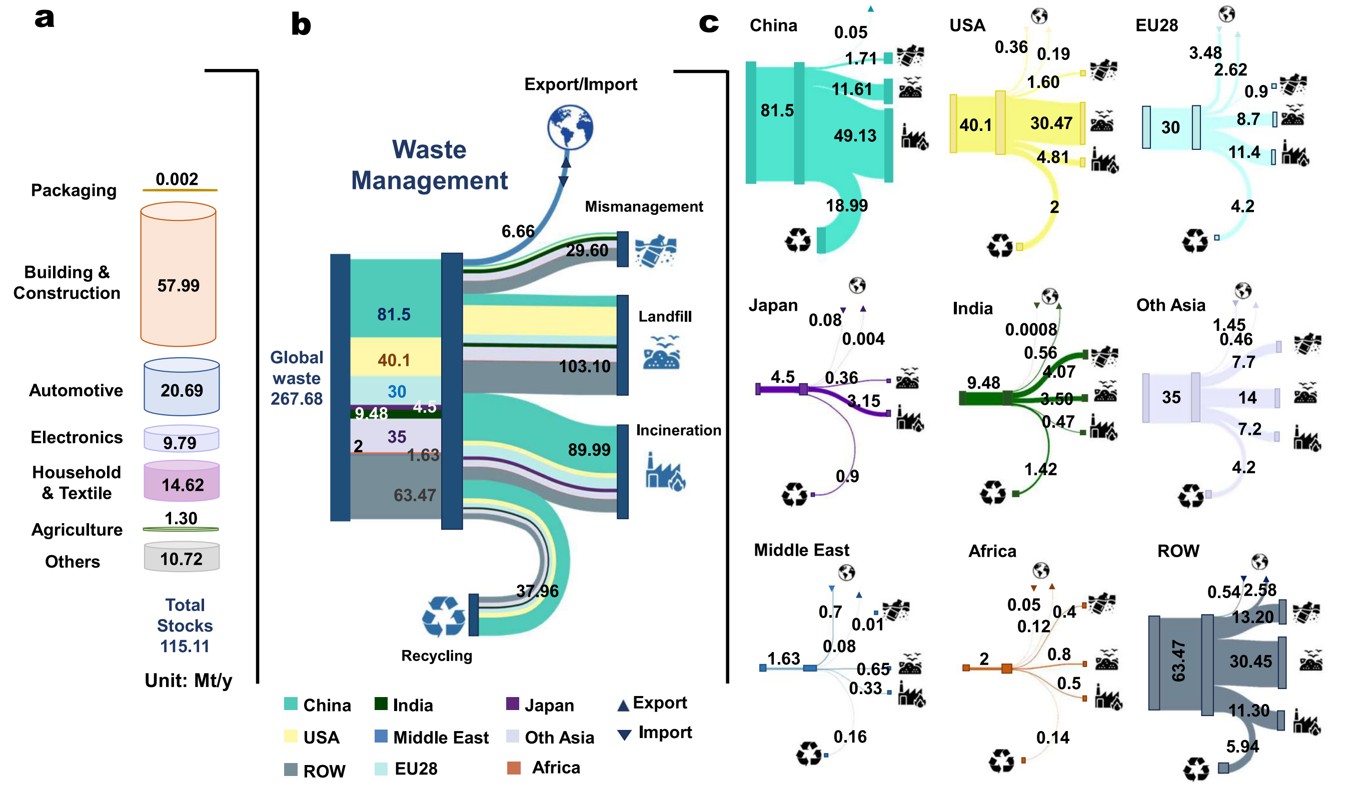

Fig. 4 | Global plastics waste generated. a Global plastics stocks per industrial sector. **b** Global waste management and end-of-life waste, breakdown per country/region. **c** Regional and national plastics waste management. Color and width of flows correspond to the countries and mass of plastic waste, respectively. All units in Mt/y units. *Note: minor discrepancies in values are due to rounding during mass balance calculations.*

recycled. The recycling rate of USA dropped significantly comparing to 2015, when plastic recycling rate reached 9%. Historically, the United States has relied on China as its largest plastic waste recipient[37]. However, when China imposed plastic waste import in 2018, plastic waste management in USA was disrupted, resulting in a significant drop of its plastic recycling trends. In addition, plastic waste in USA is estimated to grow further in the future reaching 86 Mt by 2050[35].

Discussion

The global plastics fossil-fuel source dilemma

We calculated the feedstocks for the global production of plastics in 2022. A total of 98% of the global virgin plastics was generated from fossil fuel feedstocks, with 44% from coal, 40% and 8% from petroleum and natural gas respectively. Our results are in line with previous studies[38,39] and demonstrate that in 2022 little progress has been made to reduce the reliance of fossil-fuel feedstocks for plastic production. The large amount of coal application for virgin plastics production is mainly driven by China, one of the largest consumers of coal for energy and chemical industry production, rising further environmental concerns related to plastics production. In addition, the global expansion of the petrochemical industry is resulting in a surge of oil consumption, especially from China. Around 15% of the total use of oil in China was consumed by the chemical industry[39]. The high reliance on fossil-fuel feedstocks for plastics production will further compromise the global efforts to mitigate climate change. Yet, plastics result in lower greenhouse gas emissions throughout their life cycle compared to alternative materials like metals or glass[40]. In addition, bioplastics face several challenges, including potential emissions during the production stage, environmental impacts such as land use and eutrophication, competition with food production, and economic and technological barriers for a large-scale use[41–44]. This paradox amplifies the dilemma: while plastics contribute to fossil fuel dependence and pervasive pollution issues, they simultaneously provide environmental benefits in terms of reduced emissions and energy efficiency. The lack of viable alternatives to plastics that can be implemented on a global scale presents a significant challenge in the quest for sustainable solutions. Given this paradox and the limited sustainable substitution options at a large scale, implementing circular economy principles and reducing plastic flows emerges as a crucial solution. By focusing on strategies such as recycling, reuse, and redesigning products for longevity, we can minimize the environmental impact of plastics and move towards a more sustainable future[45]. In addition, the environmental risks associated with the consumption of fossil feedstocks for the chemical industry need substantial resource efficacy and circularity intervention. This will require a shift from fossil fuel feedstocks to biogenic and recycled feedstocks, the implementation of carbon capture and sequestration technologies and the decarbonization of energy supply along the supply chain of chemical and plastic products.

The global shift of plastic trade patterns

The global trade of plastic products is a complex and expansive network, marked by significant geographic and economic interdependencies. We identified different regional hotspots in different stages of the plastic supply chain. Being that plastics are predominantly fossil-fuel based materials, oil rich countries are key exporters of refined petroleum products and petrochemicals, supporting the global supply chain for plastic production. The plastic additive industry plays a crucial role in the manufacturing sector by modifying and enhancing the properties of plastics to meet diverse application requirements. China's plastic additive market has seen rapid growth due to its expansive manufacturing sector, with a market value projected to grow at annual rate of 5.7% over the next five years[46]. Europe has stringent regulations governing the use of additives in plastic production which oblige European countries to register all additives manufactured or produced in the European Chemicals Agency[47]. Regulatory frameworks and technological advancement boost the production of innovative functional additives and the introduction of sustainable additives such biodegradable plasticizers. Yet, a large number of additives and monomers produced and traded in the global market have been identified as substances of potential concern for their toxicity based on the European Union criteria[48]. The release of toxic additives into the food from the plastics packaging and the presence of potentially toxic substances[49] further increases the need to implement stricter regulations in plastics additives production and trade. Yet, the complex nature of plastic additives, which varies widely in type, concentration, and interaction with the base polymer, poses significant challenges in accurately quantifying their flows, stocks and trade. There is a critical need for more robust statistical data and methodologies to better understand and quantify these additives. Academic institutions, regulatory bodies such as the World Customs Organization (WCO), and enterprises must collaborate closely. An accurate classification and valuation of additives are essential to ensure compliance with environmental and safety standards.

By examining the global waste trade patterns of 2022, we found that a number of developed countries, in particular EU28 are emerging as plastic waste/scrap net importer. Environmental concerns regarding plastic waste and pollution have led to international regulatory efforts and shifts toward more sustainable practices. One major factor that could have shaped the global plastics trade patterns could be the implementation of the Basel Convention Plastic Waste Amendments in 2021[50]. These amendment mandates that exporter countries conduct pretreatment of plastic waste to transform it into recycled material or clean scraps that meet specified criteria before following the Prior Informed Consent (PIC) procedure for export, especially for mixed plastic waste (entry 'Y48')[51]. These changes might result in significant GHG emission reductions and more incentives for developed countries to import clean plastic scraps, which could result in lower GHG emissions. The pretreatment process further reduces the weight of the transported materials, leading to decreased fuel consumption and emissions during transportation. In addition, high transportation costs are likely to result in future plastic waste trade being more concentrated within trade groups in the same geographical areas[52]. Overall, clean plastic waste/scrap import offers opportunities for utilizing materials that would otherwise be disposed of in landfills, promoting a more sustainable approach to waste management[53]. Importing clean plastic waste/scrap can further supplement the recycling industry of developed countries, supporting the demand for recycled plastics in manufacturing processes and reducing the environmental impact of virgin plastic production.

Policy prospective for a sustainable management of plastic waste

Global plastic waste management practices exhibit significant regional variations, reflecting differences in economic capabilities, regulatory frameworks, technological advancements, and cultural attitudes towards waste. In high-income countries, sophisticated systems are often in place, integrating advanced sorting and recycling technologies, comprehensive waste collection services, and stringent regulations. For instance, countries in the European Union have implemented extensive recycling programs and policies aimed at achieving high recycling rates and reducing landfill disposal[10]. In contrast, many developing regions struggle with limited infrastructure, leading to higher rates of improper disposal and environmental leakage.

We observed that the total waste generated (268 Mt) is lower than the amount estimated previously (320–350 Mt)[54,55], primarily due to our methodological approach we employed. We utilized a mass balance approach for the year 2022 to ensure that all inputs and outputs are accounted for, providing a more precise representation of waste generation. Unlike previous studies that often include cumulative waste from previous years, our model focuses exclusively on the waste generated within the year 2022. Furthermore, our analysis account for plastics stock (141.58 Mt) that are not included in the waste stream for 2022 because they are still in use. These stocks are retained within various sectors where the lifespan of the plastic material extends beyond one year (Fig. 4a, Supplementary Table 6).

The global recycling rate remain stagnant showing little progress from previous estimates[1], accounting of 9% of primary production. Only 38 Mt (out of the 75 Mt sorted and collected) is recycled. Several challenges contribute to the limited progress in enhancing global recycling practices. One major issue is the vast diversity and complexity of plastic materials, which include various types, grades, and additives[56]. This complexity limits the efficient sorting and processing of plastic waste, as different types often require separate processing to maintain quality. Additionally, contamination of plastic waste with food residues, labels, and other impurities further undermines recycling efforts, as these contaminants can degrade the quality of recycled materials and create operational challenges in recycling facilities[57]. Economic factors also play a crucial role; the market conditions surrounding the cost of virgin plastic are often more favorable than those for recycled plastic, primarily due to fluctuating oil prices[58]. This economic barrier discourages investment in recycling infrastructure and technology, perpetuating the cycle of low recycling rates. Furthermore, the lack of mechanism promoting product design for higher resources efficiency means many products are difficult to be dismantled or recycled effectively, or even cannot be recycled due to the additives that already been prohibited for use in products. Eco-design and standardization at the design stage are critical factors for enhancing the recycling rate of plastics. By integrating sustainability principles into the design process, manufacturers can create products that are not only functional but also conducive to efficient recycling. This includes choosing recyclable materials, minimizing the use of additives that can interfere with recycling processes, and ensuring that plastics products can be easily recycled at the end of their life cycle[59,60].

Our findings show that USA, despite advances in recycling technologies has one of the lowest plastics recycling rates (5%), facing significant challenges in its plastic waste management[17,35]. Increasing plastics recycling rate, integrating circular policies and limiting landfill could benefit USA waste management practices. European countries set an ambitious target to achieve 77% of collection of plastics bottles by 2025, and 90% by 2029 and incorporating 25% of recycled plastic in PET beverage bottles from 2025[61]. Similarly, in 2019 Japan adopted the Plastic Resource Circulation Strategy with the aim to implement circular economy strategies in the plastic sector by achieving 60% of plastic recycling by 2030 and reduce single-use plastic emissions by 25% by 2030. The country set an ambitious target to maximize the production of bioplastic by 2030, reaching approximately 2 Mt[62]. China formulated official measures for the management of agricultural films in 2020 with the aim to limit the environmental impact of mulch film on soil and promote their reuse and recycling[63]. Agricultural film pollution is indeed a major factor of the degradation of soil in rural areas and a growing source of contamination to the environment and human health[64]. In response, the Chinese government set as a target to achieve 85% of collection rate for agricultural films by 2025[65].

Based on our findings Japan, China and EU28 have among the highest incinerations rates for plastic waste accounting for 70%, 60% and 38% respectively. Europe's high plastic incineration rate is attributed to the continent's focus on energy recovery and waste-to-energy initiatives, serving as a mean to reduce landfill dependence and mitigate environmental impacts associated with plastic waste[66]. In Japan, incineration plays a significant role in waste management due to limited landfill space, strict regulations on waste disposal, and advanced technologies that allow for efficient energy generation from waste incineration[67,68]. China's rapidly growing industrial sector and escalating waste generation have led to an historical increase of incineration rate for solid waste management as an alternative to landfill waste practices[69]. By September 2022, there were 811 incineration facilities to energy recovery in China and this number is expected to grow further[70]. Controlled incineration can reduce waste and produce energy, yet it also poses environmental risks resulting in the release of harmful pollutants that require advanced technologies and stringent regulatory framework so that the negatives impacts could be limited.

Plastic production, consumption, and waste disposal in Africa present unique challenges and opportunities. While the region has seen a significant increase in plastic production and consumption driven by economic growth and urbanization, waste management systems in Africa face unprecedented challenges. Inadequate waste collection, improper disposal practices, and limited recycling infrastructure exacerbate environmental pollution, human health risks, and resource depletion in the continent[71]. Despite challenges, there is a growing recognition of the need for sustainable plastic waste management solutions[72]. We found that the average plastic recycling rate in Africa is around 7%, however this rate might be higher when considering informal recycling, conducted by waste pickers or informal recyclers. Informal plastic recycling in the African continent and in other developing countries plays a critical role in waste management, offering both benefits and posing significant challenges. Such waste management practices are often characterized by small-scale, unregulated operations and they provide livelihoods for millions of people[73–75]. Despite their economic and environmental benefits, informal recycling practices often occur under hazardous conditions that expose workers to health risks from toxins and pollutants[76]. Addressing these issues requires integrated approaches that enhance the capabilities of the informal sector through improved technologies, supportive policies, and international cooperation, aiming to protect both communities and the environment. Enhancing the synergies between informal and formal recycling systems can streamline plastic waste management and promote sustainability in developing countries.

Circular economy principles for a global sustainable management of plastics

Circular economy principles offer a strategic approach to managing plastic material flow and mitigating global pollution[77,78], aligning with the findings of our study: (i) Reuse: reusing plastic products extends their life cycle, keeping them in circulation and out of the waste stream. Encouraging the reuse of plastic containers and bags can significantly reduce the volume of plastic waste generated, especially for packaging, supporting more sustainable practices in plastic management. This principle further advocates for the avoidance of unnecessary plastic production and consumption. Banning the production of unnecessary single-use plastics, the demand for plastic feedstocks can be significantly reduced. This aligns with the need to decrease reliance on fossil fuels, which are the primary source of plastic production. (ii) Reduce: reducing the overall use of plastics and the resources required for their production is essential. Efficient product design that minimizes material use, combined with consumer education to promote durable and reusable alternatives, can help achieve this goal. This reduction can mitigate the environmental impacts associated with plastic production. (iv) Recycle: recycling is a critical strategy for managing plastic waste. Enhancing plastics recycling will promote the reduction of demand for virgin plastic production and lowering greenhouse gas emissions. However, the global recycling rate remains stagnant, highlighting the need for improved recycling infrastructure and processes. (v) Restore: this principle involves restoring and regenerating natural systems that have been degraded by plastic pollution. Efforts to clean up plastic waste from the environment and restore ecosystems are crucial for mitigating the pervasive environmental impacts of plastic pollution. Implementing these R principles can significantly reduce the environmental impact of plastic production and consumption, aligning with the urgent need for sustainable solutions to address global plastic pollution.

Study limitations

While our study provides valuable insights into the global plastic flows, stocks and waste disposal, it is important to acknowledge its limitations. Variations in data collection practices across different regions may introduce inaccuracies. Collaborating with government and academic institutions to standardize data collection methods and improve data quality would be beneficial for future research. This would ensure that the data used is more consistent and reliable, allowing for more accurate comparisons and analyses. Furthermore, a global accounting of plastics is inherently complex due to the intricate nature of trade flows and the diverse range of plastic types and additives involved. While our study provides a comprehensive overview of global plastic flows, stocks, and waste disposal, it is limited in its ability to offer detailed analyses of specific additives' production and disposal. We have focused primarily on the most commonly used and traded additives, which may not capture the full spectrum of plastic additives in circulation. In future research, we aim to delve into the multitude of plastics and additives to provide a more comprehensive understanding of their individual flows and impacts. Additionally, our study does not account for the significant role of informal recycling systems, which play a crucial part in the global plastic waste management. This lack of inclusion may affect the accuracy of the global recycling rate, which might be higher if informal waste recycling is accounted. Future research should consider integrating data from informal recycling networks to gain a more holistic view of plastic waste management and recycling practices worldwide.

Conclusions

In this study we uncover the global flows and stocks for the year 2022 by providing a first global trade-linked material flow analysis with breakdown per plastic type along the whole supply chain from feedstocks to waste management disposal. We estimate and illustrate the global waste management to reveal the fates of plastic waste and connect the missing dots dominating the complex world of plastics. We further map the global plastic trade along the whole supply chain. Our findings suggest that plastics are subject to geographical concentration. Plastic feedstocks, predominantly derived from fossil-fuel materials, and the primary production of plastics are intensely concentrated in countries endowed with abundant oil resources and possessing advanced petrochemical industries. The processing and manufacturing of plastics are predominantly concentrated in regions characterized by large-scale manufacturing capabilities. Notably, China alone accounts for 40% of the global output of final plastic products. This geographical concentration has significant implications for global trade dynamics, environmental policies, and sustainable practices, much like the concentration observed in the mining and processing of metals and minerals.

Results from the global plastics waste management fates show that incineration is emerging as the most practiced method for managing plastic waste, accounting for 34% of global plastic waste treatment. This shift is occurring alongside a notable decline in plastic landfill disposal, which constitutes 40% of plastic waste management methods globally. Despite these developments, the global recycling rate remained stagnant at 9% (38 Mt), reflecting little improvement from previous years. Moreover, the dynamics of the global plastics trade are undergoing significant changes, with developed countries becoming prominent importers of plastic waste and scrap. EU28, in particular, has emerged as a net importer, accounting for 52% of global plastic waste imports.

Understanding and quantifying the global movement and accumulation of plastics are vital for guiding policy decisions, implementing effective management strategies, and monitoring progress towards environmental sustainability goals. The insights gained from our findings provide valuable data and evidence to inform academia and policymakers on the global plastic flows, enabling countries and stakeholders to collaborate effectively in addressing this pressing global environmental challenge. Accounting for global flows and stocks of plastic is not only a scientific necessity but also a practical tool for policymakers to design evidence-based policies and regulations that drive progress towards a cleaner, healthier planet for current and future generations.

Methods

Research framework

The system diagram of the global plastic cycles is illustrated in Fig. 5. We cover the following plastics cycles: virgin resin production, product processing, product manufacturing, use and waste management. The global plastic trade linked MFA is based on the year 2022 because it provides the most comprehensive and recent data available, allowing for a detailed analysis of current trade dynamics and patterns. Additionally, an analysis of the year 2022 reflects the post-pandemic recovery and the latest trends in plastic production and trade, offering valuable insights for understanding the current state of the global plastic industry.

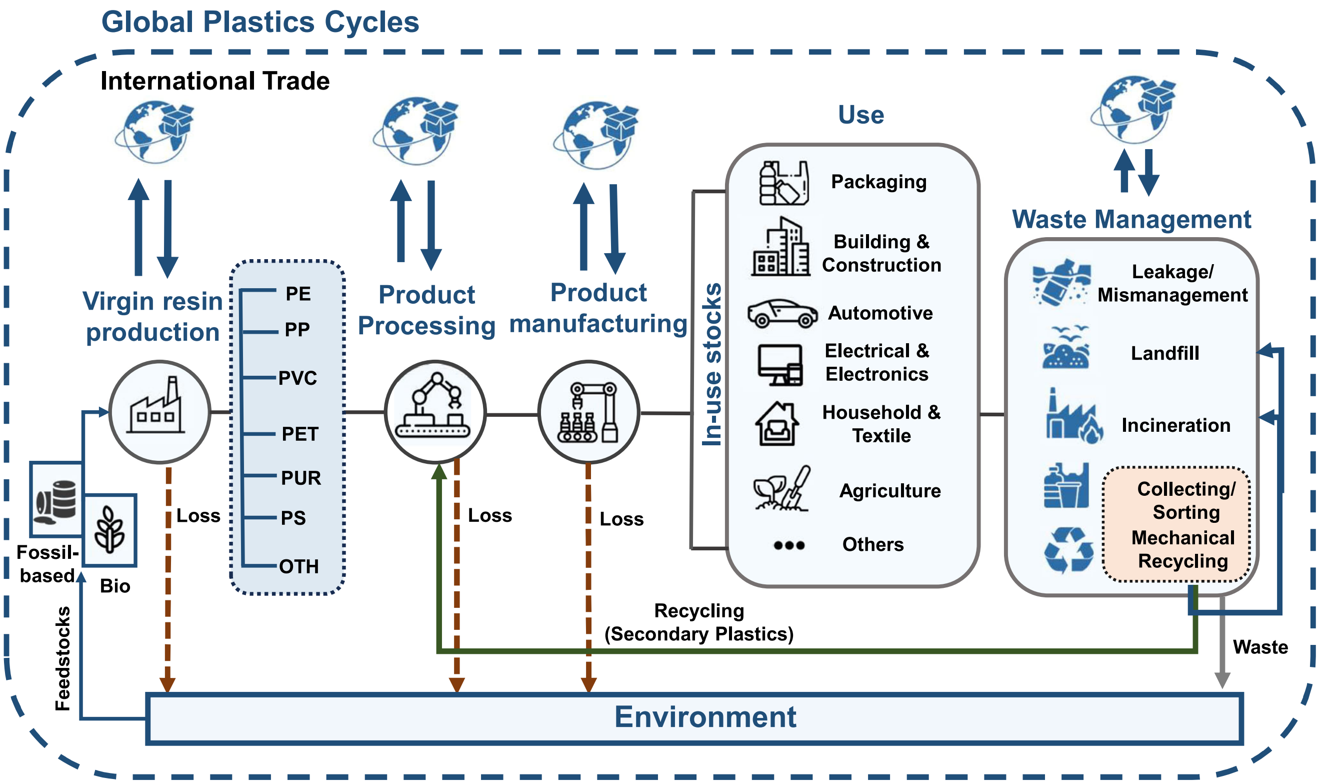

Fig. 5 | Global plastic cycles system diagram. The system includes selected polymers: Poly(ethylene) (PE), Poly(propylene) (PP), Poly(vinyl chloride) (PVC), Poly(ethylene terephthalate) (PET), Poly(urethane) (PUR), Poly(styrene) (PS) and others (OTH) and analyzes their cycles from feedstocks production to waste management.

The system diagram involves different stages along the supply chain; (i) virgin resin production: this stage involves the production of resins from natural and synthetic resources such as petrochemical feedstocks. (ii) processing and manufacturing: this stage involves the transformation of virgin resins into semi-finished products and final plastic-containing products. The plastic types that will be considered are Poly(ethylene) (PE), Poly(propylene) (PP), Poly(vinyl chloride) (PVC), Poly(ethylene terephthalate) (PET), Poly(urethane) (PUR), Poly(styrene) (PS) and others (OTH); (iii) use stage: the consumption of plastic products is categorized into sectors such as packaging, building and construction, automotive, electrical and electronics, household and textile, agriculture, and others. This selection of these sectors is based on extensive previous literature[1,8,9,25]. These sectors are identified as major consumers of plastics due to their significant contribution to overall plastic consumption and environmental impacts. A description of the plastics products comprising each of the selected plastics use category is provided in Supplementary Table 2; (iv) waste management: the final stage involves the management of plastic waste, including mismanagement, landfilling, incineration, and mechanical recycling. The recycling of plastics is considered as a source of secondary plastics, which reenter the product processing stage.

Global plastic trade-linked material flow analysis

The global accounting of plastics flows follows the mass balance principle[7], ensuring that the inputs and outputs of plastics are accurately tracked across different countries and regions. This involves calculating domestic flows, trade flows, loss flows, and recycling flows using the following Eq. (1):

$$\mathrm{F^{input}_{Px,i,j} + F^{import}_{Px,i,j} + F^{recycling}_{Px,i,j} = F^{output}_{Px,i,j} + F^{export}_{Px,i,j} + F^{loss}_{Px,i,j} + \Delta S_{Px,i,j}} \tag{1}$$

where i denotes the ith life stage of plastics material type (Px), j denotes the jth country covered in this study, $F^{input}_{Px,i,j}$ is the Px contained in domestic feed-in flows from ith stage in country j, $F^{import}_{Px,i,j}$ is the Px embedded in import of plastics contained products from ith stage in country j, $F^{recycling}_{Px,i,j}$ is the plastics recycling in the ith stage in country j, $F^{output}_{Px,i,j}$ is plastics contained in products consumed by domestic demand from ith stage in country j, $F^{export}_{Px,i,j}$ is exported plastics embodied in plastics contained products from the ith stage in country j, $F^{loss}_{Px,i,j}$ is plastics loss during the production of i stage in country j, $\Delta S_{CO,i,j}$ is the plastics net addition to the stock of the ith stage in country j.

Plastics inflows and outflows in the use stage will be calculated based on the delay model[79]. The outflow in a certain year is equal to the inflow of the past and is calculated by using Eqs. (2) and (3):

$$\mathrm{f^t_{i,inflow} = F^t_{i,inflow} \times P^t_i} \tag{2}$$

$$\mathrm{f^t_{i,outflow} = F^t_{i,inflow} \times P^{t-n}_i} \tag{3}$$

where $\mathrm{f^t_{i,inflow}}$ and $\mathrm{f^t_{i,outflow}}$ refer to the plastics inflow and outflow of s industry i in year t, whereas $\mathrm{F^t_{i,inflow}}$ and $\mathrm{F^t_{i,outflow}}$ refer to the total plastics inflows in the use stage in year t and year t-n, respectively; $\mathrm{P^t_i}$ and $\mathrm{P^{t-n}_i}$ refer to the production of plastics used in end-use category in year t and year t-n, respectively.

The plastics consumption is assumed based on the known production and trade data. Thus, the relationship among plastics production, net trade and consumption follows the principles of mass balance. Consumption of plastics at national and regional is calculated based on Eq. (4):

$$C_{S_s t_n} = \sum_{p=1}^{7} (P_{p,t_n} + Ni_{p,t_n} + Wi_{p,t_n}) * r_{p,s,t_n} + Rw_{p,s,t_n} + NP_{p,t_n} \tag{4}$$

where C_{S_s} is the consumption of plastics in sector s at time t_n, P_{p,t_n} is the production of plastics (resin or fiber) at time t_n, Ni_p represents the net import of plastics at time t_n, Wi_{p,t_n} is the waste import of plastics at time t_n, r_{p,s,t_n} is the ratio of resin or fiber in sector s at time t_n, Rw_{p,s,t_n} is the recycled plastic waste at time t_n and, NP_{p,t_n} is the net import of finished products in sector s.

Finally, all the yield ratios and losses along the different stages are calculated using transfer coefficients based on industrial statistics and their related estimates. These include waste from virgin resin production, waste from semi-manufacturing processes, waste from product manufacturing and finally waste and losses from waste management process. While these losses are quantified as entering the environment, their specific environmental pathways (aquatic or terrestrial) and ultimate destinations are not modeled in our study.

Plastics stocks accounting

In this study, we quantify plastic stocks using the MFA approach based on mass balance principles. The calculation of plastic stocks involves assessing the inputs, outputs, and accumulation of materials within a defined system boundary over a given time period.

The plastics stock for a given system and timeframe is determined using Eq. (5):

$$S_t = P_t + Im_t - Ex_t - W_t \tag{5}$$

where S_t is the stock of plastics at time (t), P_t is the total production of plastics, Im_t represents the total imports of plastics into the system at time (t), Ex_t denotes the total exports of plastics from the system, and W_t is the total waste generation within the system.

Global and regional waste accounting

The total waste generated in 2022 (w(t)) was calculated based on Eq. (6):

$$W(t) = \sum_{i=1}^{7}\sum_{j=1}^{2022} P_i(t-j).\, LTD_i(j) \tag{6}$$

where $\sum_{i=1}^{7}\sum_{j=1}^{2022} x$ indicates the sum of the plastics application sectors in year 2022, $P_i(t-j)$ the amount of primary plastics produced in year *(t)* and used in sector *(j)* and $LTD_i(j)$ indicates the fraction of plastics in industrial use sector (i) used for j years.

Data collection

Data collection for each country and region are collected from various sources, including national statistics, industry reports, and international databases such as the UN Comtrade. The use of Harmonized System (HS) codes covers a broad range of plastic materials, including feedstocks, additives, primary plastics, intermediate plastics, final products, and waste. We have totally collected 566 HS codes (37 HS codes for plastics feedstocks, 80 for plastics additives, 67 for primary plastics in polymers and monomers forms, 115 for intermediate plastics, 263 for final plastics, and 4 for plastics waste. HS codes of plastic materials and products are further discussed in Supplementary Note 2. To ensure accuracy, trade data are cross-checked with regional trade reports and industry publications, resolving discrepancies by consulting additional sources or adjusting data based on known trade patterns. Industry reports and surveys offer insights into plastic production capacities, manufacturing processes, and consumption patterns. The product lifetime might vary significantly across regions, therefore based on previous published literature we have considered distributions and conducted sensitivity analysis (Supplementary Table 6). The global and regional waste management ratios are provided in Supplementary Tables 7 and 8 respectively.

Sensitivity analysis

The sensitivity analysis is designed to assess the impact of variations in key parameters on the outcomes of the global trade-linked MFA model. Parameters such as global and regional plastics production rates, consumption rates, waste generation metrics, and plastics product lifespans are systematically varied to evaluate their influence on the model's results. For global and regional plastics production, variations of ±10% and ±20% are applied. Consumption rates are adjusted by ±5% and ±10%, while plastic waste generation rates are altered by ±5 kg and ±10 kg per capita. The plastic content of various products is varied by ±2% for low uncertainties and by ±10% for high uncertainties. Uncertainties in trade data are assessed by comparing reported import and export figures between trading partners. Discrepancies are quantified as a percentage of the total trade volume, providing insights into potential data inconsistencies and informing adjustments to trade estimates (Supplementary Note 3, Supplementary Table 9).

Data availability

All data needed to evaluate the conclusions in the paper are present in the main text and/or in the Supplementary Materials. The dataset used in this study is freely available in the open-access repository Zenodo (https://doi.org/10.5281/zenodo.14906505). Any further information regarding this paper can be obtained upon request from the authors.

Received: 8 September 2024; Accepted: 26 February 2025;
Published online: 10 April 2025

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Acknowledgements

This study is financially supported by "National Natural Science Foundation of China" (72274105, W2433174).

Author contributions

K.H. conceived the study, conducted data collection and technical analysis. J.L research conceptualization and design, result analysis. Q.T contributed to methodology development, data collection and performed the analysis.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s43247-025-02169-5.

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Peer review information *Communications Earth and Environment* thanks Willi Haas and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Somaparna Ghosh [A peer review file is available].